# Mamba Collaborative Implicit Neural Representation for Hyperspectral and Multispectral Remote Sensing Image Fusion

Chunyu Zhu, Shangqi Deng, Xuan Song, *Member, IEEE*, Yachao Li, *Member, IEEE*, and Qi Wang

***Abstract*— Hyperspectral remote sensing images (HSIs) capture detailed spectral characteristics of features, while multispectral remote sensing images (MSIs) provide clear spatial distribution. Fusing these two types of images can enhance feature identification and classification accuracy. Current deep learning algorithms achieve high fusion quality but struggle with balancing global effective perception and lightweight computation. Moreover, these algorithms typically discretely handle data mapping, which contrasts with the continuous nature of the world. Recently, the Mamba has shown significant potential for complex long-range modeling, addressing the computational complexity of global perception. Concurrently, implicit neural representation (INR) offers high-quality solutions for continuous domain modeling. To this end, this study introduces a novel network architecture that combines Mamba and INR, termed the Mamba cooperative INR fusion network (MCIFNet). MCIFNet effectively captures global image information and generates fused images in a continuous domain through point-to-point processing. The network comprises two main units: potential space projection and semantic extraction and fusion. The potential space projection unit performs shallow encoding of hyperspectral and MSIs, mapping them to a latent feature space. The semantic extraction and fusion unit (SEFU) uses scale adaptive residual state spatial and implicit spatial-spectral fusion (ISSF) modules to extract deep features from the bimodal images, generating fused images point-by-point. A series of fusion experiments with 4×, 8×, and 16× scale factors demonstrate that MCIFNet surpasses popular algorithms in both spatial detail and spectral information reconstruction, while also providing more lightweight performance. The code for MCIFNet will be shared on https://github.com/chunyuzhu/MCIFNet.**

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***Index Terms*— Hyperspectral and multispectral image fusion, implicit neural representation (INR), Mamba, Mamba cooperative INR fusion network (MCIFNet), state-space model (SSM).**

## I. Introduction

HYPERSPECTRAL and multispectral remote sensing image fusion (HMIF) is a commonly used remote sensing data processing technique aiming to enhance the spatial resolution of hyperspectral remote sensing images (HSIs) through the high spatial resolution property of multispectral remote sensing images (MSIs) [1]. HSI is rich in spectral information, which enables accurate identification of objects or scenes [2]. However, due to its low spatial resolution, it restricts the monitoring of wide-area features and the extraction of fine details [3], [4]. In contrast, MSI has high spatial resolution, but its low spectral resolution makes it difficult to perform fine spectral analysis and feature differentiation [5]. With HMIF technology, the advantages of these two types of images can be more effectively combined to improve the range of data applications [6], such as land classification [7] and target recognition [8].

HMIF covers two mainstream methods: traditional methods and deep learning [9], [10]. Traditional methods are usually based on imaging models, which often have nonunique and unstable solutions due to the ill-defined nature of the problem, and require the introduction of the prior terms to constrain the optimization process [11]. The commonly used priors include sparse prior [12], spatial nonlocal similarity prior [13], low-rank prior [14] etc., which guide the optimization direction of the fusion algorithms to some extent. The above artificially defined priors are often insufficient to accurately capture effective mappings between the data, leading to shortcomings in the fusion accuracy of traditional methods [15]. In contrast, deep learning methods are able to better perceive the complex nonlinear mappings between the data by virtue of their powerful fitting capabilities [16], thus improving fusion accuracy and effectiveness [17]. Currently, HMIF often combines the local perception of the convolutional neural network (CNN) and the global interaction of transformer [18] to achieve high-quality fusion. However, due to the inherent characteristics of the transformer's self-attention mechanism, the computational complexity of the algorithm is quadratically related to the input, making it difficult for the algorithm to balance the contradiction between accuracy and computational efficiency.

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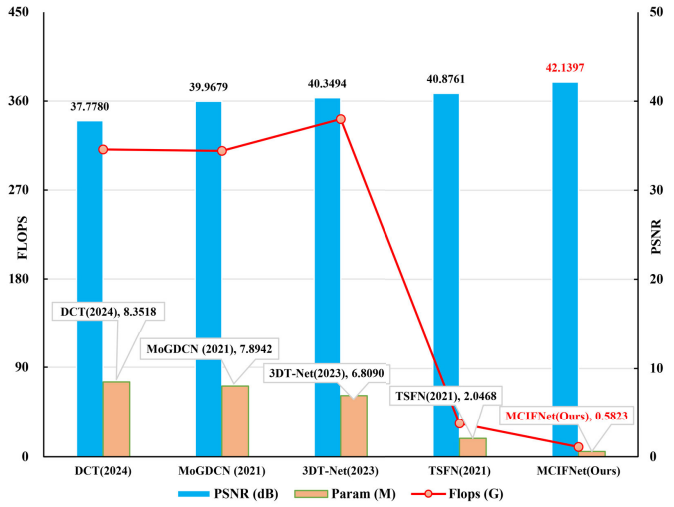



Fig. 1. Comparison of PSNR and FLOPs between MCIFNet and other deep learning algorithms, at a scale factor of 4× on the PaviaC dataset with fusion images of size 128 × 128, clearly indicates that MCIFNet achieves superior fusion accuracy while being the lightest weight.

In addition, the existing algorithms usually adopt primary methods such as addition or concatenation for feature fusion, which ignores modal differences and leads to information redundancy to a certain extent. Finally, most of the algorithms still perceive images in a discrete way, which has a certain gap with the continuous perception in the real world.

Recently, Mamba [19] integrates structured state-space models (S4) [20], which has become an indispensable core supporting technology in several vision tasks. In the Mamba framework, the discrete state-space equations are presented in a recursive form, and with the help of structured reparameterization techniques, the long-range interaction of features can be perceived effectively [21]. Compared with the traditional transformer, Mamba has a faster inference speed due to the linear time complexity feature. In addition, implicit neural representation (INR) is a novel representation method that uses continuous differentiable functions to map coordinates to specific signals, which is able to transform discrete domains to continuous domains, and possesses a stronger representation capability than previous discrete perceptions. In fact, the Mamba model may have limitations in capturing local information in a fine-grained manner when constructing the global view as it focuses on the overall scene understanding. The INR, in contrast, is able to achieve finer information enhancement at the detail level by directly enhancing the feature representation using a point-to-point functional form, which demonstrates superior performance in the identification and representation of local features. Therefore, the combination of Mamba and INR may bring new potential to the field of HMIF, by making full use of the advantages of each other, which can grasp the overall structure of the scene and accurately deal with the local details, and achieve greater efficiency and comprehensiveness of feature perception.

In this study, we propose a novel Mamba collaborative INR fusion network (MCIFNet) that combines the advantages of Mamba and INR to alleviate the current contradiction between fusion accuracy and degree of lightweight, which is reflected in Fig. 1. Specifically, MCIFNet consists of two key units: the latent space projection unit (LSPU) and the semantic extraction and fusion unit (SEFU). First, LSPU conducts shallow encoding of HSI and MSI and projects them into the potential space for subsequent feature extraction. The SEFU module comprises two key components: the scale adaptive residual state spatial block (SARSSB) and the implicit spatial-spectral fusion (ISSF). Initially, SARSSB leverages its robust representational capabilities to effectively extract deep features from the images, thereby enhancing the richness of feature representation. Subsequently, ISSF uses the deep features of the MSI to guide the implicit-fusion-based upsampling of HSI deep features, achieving ISSF and generating high spatial resolution HSIs (HR-HSI). The experimental results on the DFC2018, Chikusei, PaiviaC, and PaviaU datasets show that MCIFNet outperforms current popular algorithms on ×4, ×8, and ×16 scale factor fusion, effectively balancing the contradiction between fusion accuracy and model lightweight. The specific contributions of this study are as follows.

1) An efficient HMIF algorithm MCIFNet is proposed to effectively sense local and global information.
2) We have designed SEFU to alleviate the insensitivity of Mamba to local information.
3) Based on ISSF, we effectively implement cross-modal feature point-to-point fusion and achieve better results on several datasets.

This study is organized as follows: Section II introduces related work, including the review of traditional and deep learning algorithms for HMIF and the development of Mamba; Section III describes the network structure of MCIFNet and the loss function used in this article; Section IV analyses the experimental results of MCIFNet compared with the comparative popular algorithms and performs ablation experiments; and Section V concludes the whole article and looks forward to the future work.

## II. Related Work

### A. Pansharpening-Based HMIF

Pansharpening was originally applied to the fusion of MSI and panchromatic images (PANs), which can be regarded as a special case of HMIF [22], and mainly involves two techniques, component substitution (CS) and multiresolution analysis (MRA). The CS method separates the spatial components of the low spatial resolution image through specific transformations and subsequently replaces these components with PAN, which undergoes an inverse transformation to generate the fused image. The typical CS methods include PCA [23], GS [24], and GSA [25], etc. The MRA performs multiresolution decomposition of the images, then embeds the high-frequency information of the PAN into the MSI, and finally obtains the fused image by inverse transformation. Gomez et al. [26] proposed a wavelet-fusion-based method which performs wavelet decomposition of HSI and MSI, interpolates HSI based on the high spatial resolution property of MSI, and finally obtains HR-HSI by inverse transformation. Chen et al. [27] proposed an HMIF framework which achieves fusion by grouping HSIs based on their spectral range and then applying pansharpening to each grouping. Although panchromatic image sharpening-based methods are simple in design

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and easy to implement, the fusion quality usually fails to meet the requirements of practical applications when facing fusion tasks with higher scale factors [28].

### B. Decomposition-Based HMIF

Matrix decomposition decomposes the 3-D image into the product of the 2-D spectral basis and the coefficient matrix. The spectral basis and coefficient matrices of HR-HSI are estimated by solving the optimization problem to generate the fused image. The modeling of spectral bases can generally be categorized into sparse representation and low-rank representation. The typical sparse representation such as Kawakami et al. [29] applied the unmixing algorithm to HSI for the estimation of the spectral base representation, and then combined this representation with MSI to produce HR-HSI. Han et al. [30] proposed a self-similarity constrained sparse representation for MSI exploring self-similarity using global-structural groups and creating local-spectral super-pixels, combining this self-similarity with the spectral dictionary of HSI to achieve fusion. Although sparse representation is able to reduce redundancy between spectra, its ability to compress spectral dimensions is relatively weak compared with low-rank representation [28]. Zhou et al. [31] combined linear spectral decomposition with local low-rank features by segmenting the images into small patches and processing them using the local features, and then extracting the abundance and end elements of MSI and HSI, respectively, through coupled spectral unmixing to achieve HMIF. Xue et al. [32] considered the spatial–spectral subspace low-rank relationship between MSI and HSI and the target image and proposed a new subspace clustering method of structured sparse low-rank representation (SSLRR), which is used to learn the spatial features of MSI and the spectral features of HSI, and then solves the optimization problem to generate the fused image. Since images are 3-D data, the matrix-decomposition-based fusion to force them to be represented as 3-D matrices somewhat destroys the structural information. Tensor decomposition decomposes the images into third-order tensors, which is more advantageous in preserving structural information compared with matrix decomposition. Common decompositions include Tucker decomposition, singular value decomposition, and CP decomposition. Dian et al. [33] proposed NLSTF for the first time by applying Tucker decomposition, which exploits the nonlocal spatial similarity by grouping the images, and in each group, the same dictionary with sparse prior is used for decomposition, and the HMIF is achieved by learning the dictionaries of the three modalities and estimating the sparse core tensor for each cube. Dian and Li [34] exploited the spectral correlation and nonlocal similarity in HR-HSI, expressed it in the form of spectral subspace and coefficient multiplication, learned the spectral subspace of HSI through singular value decomposition, and implemented HMIF using low-tensor multirank a priori estimation coefficients. Kanatsoulis et al. [35] introduced a coupled tensor decomposition framework, captured the correlation between HSI and MSI in different dimensions by CP decomposition, and then implemented image fusion using least-squares estimation. Decomposition-based methods require the computation of a large number of parameters during the fusion process leading to slower inference, and the accuracy of fusion is often unsatisfactory for larger scale factors.

### C. Deep-Learning-Based HMIF

Unsupervised deep learning, whose input is the data to be fused and the fusion results can be directly output through the optimization iterations [36], consists of both the coding–decoding and generative-constraint approaches. The coding–decoding approach relies on the autoencoder to learn the abundance and end-elements of the images and thus reconstruct the target image. The HyCoNet proposed by Zhenget al. [37] consists of three coupled autoencoder networks, aiming to use a linear spectral unmixing model to unmix HSI and MSI into end-element and abundance information, and then achieve fusion. Similarly, Yao et al. [38] developed a coupled unmixing algorithm called CUCaNet, which adaptively learned the spectral response and point diffusion function through coupled spectral unmixing techniques and achieved effective spatial–spectral information transfer by designing a cross-attention module. The generative-constraint aspect relies on external information or degradation model constraints to achieve image fusion using deep neural networks. Zhang et al. [39] used deep networks to model the degradation process and used the deep image prior as a regularization term for image fusion. Li et al. [40] proposed a method called UDLAN that uses a three-stage training strategy. In the first two phases, the method improves the performance by learning the degradation process, while in the third phase, global fine-tuning is performed to generate more accurate high-resolution HSI images. Unsupervised deep learning has a clear physical meaning; however, the fusion is less efficient due to the large number of parameters that need to be computed during the fusion process. Supervised deep learning methods are able to fit the nonlinear mapping between the image pairs to be fused to HR-HSI directly by the network without relying on any prior. Zhu et al. [41] proposed MGDIN, which improves the quality of fusion using the residual multiscale convolution and the global context block and proposes a new loss function by combining content, spectral, and edge information. Wang et al. [42] proposed TSFN based on the depth prior method, which uses the depth prior as a regularization term integrated into the objective function and the Euclidean distance squared as a regularization factor and uses the minimum distance criterion to achieve high-quality fusion by automatically selecting the regularization parameter using the golden section search strategy. The above-mentioned methods use local optimization of CNNs to achieve HMIF, and these methods have certain shortcomings in long-range perception [43]. To enhance the global interaction between image features, researchers have started to leverage the transformer, which has performed well in long-range perception, to achieve more effective HMIF. Fusformer [44] is the first application of transformer to HMIF, which takes advantage of the rich spectral information in HSI and injects residual information into the HSI image by estimating the residual information through the transformer model. PSRT [45] uses a method called "shuffle and reshuffle strategy," which combines the windowed

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multihead self-attention (MSA) mechanism to expand the correlation region between features. In addition, to enhance the interpretability of the network, some unfolding-based methods have emerged, such as MoGDCN [46] and 3DT-Net [47], which take advantage of the fitting ability of deep learning and the interpretability of model optimization to achieve HMIF. Although supervised deep learning takes some time during training, it has stability, high accuracy, and fast inference advantages.

### D. From SSM to Mamba

The state-space model (SSM) [20] originated from the field of classical control theory and is capable of solving the long-range dependent distance problem efficiently with linear computational complexity [48], which has a similar goal to the hidden Markov model (HMM) of modeling discrete data sequences over time, with the difference that the SSM uses continuous latent variables. Compared with methods relying on the attention mechanism, SSM adopts the cyclic scanning approach to integrate the hidden states step by step and captures the contextual information of the input sequence at one time, which has the advantage of sequential and efficient processing. The Mamba introduces an input adaptive selection mechanism based on SSM, which can adaptively optimize the information processing process according to the input information by dynamically adjusting the state-space parameters, and its computational efficiency is superior compared with the transformer model of the same size. With the development of Vision Mamba, this framework is further extended to the vision domain and has achieved excellent results in tasks such as image restoration [49], segmentation [50], and classification [51], showing its potential for cross-domain applications. Nevertheless, Mamba is still in the preliminary stages of development, focusing mainly on modeling discrete data domains. Applying Mamba with INR to HMIF may be a potential research direction that promises to integrate the advantages of discrete and continuous data representation.

## III. Methodology

This section provides an exhaustive description of the MCIFNet framework designed specifically for HMIF. First, we present the process of the SSM, followed by a demonstration of the local enhanced spectral and attention module (LESAM) that enhances the local feature perception of Mamba, then we present the designed MCIFNet framework and its component units, and finally, describe the adopted loss function.

### A. Preliminaries

*1) State-Space Model:* The SSM can be generally represented as

$$\dot{x}(t) = A(t)x(t) + B(t)U(t) \tag{1}$$
$$y(t) = C(t)x(t) + D(t)U(t) \tag{2}$$

where $x(t) \in \mathbb{R}^n$, $y(t) \in \mathbb{R}^q$, and $U(t) \in \mathbb{R}^p$ represent the state, output, and input vectors, respectively. $A(t) \in \mathbb{R}^{n\times n}$, $B(t) \in \mathbb{R}^{n\times p}$, $C(t) \in \mathbb{R}^{q\times n}$, and $D \in \mathbb{R}^{q\times p}$ denote the state matrix, input matrix, output matrix, and feedforward matrix, respectively. When there is no direct feedthrough in the system model, $D(t)$ is a zero matrix, thus we obtain the following equation:

$$\dot{x}(t) = A(t)x(t) + B(t)U(t) \tag{3}$$
$$y(t) = C(t)x(t). \tag{4}$$

Since the original system matrix is continuous, we need to discretize it for computational purposes. Here, we use zero-order hold (ZOH) for discretization, as follows:

$$x_t = \bar{A}x_{t-1} + \bar{B}U_t \tag{5}$$
$$y_c = Cx_t \tag{6}$$

where $\bar{A}$ and $\bar{B}$ are defined as

$$\bar{A} = \exp(\Delta A) \tag{7}$$
$$\bar{B} = (A)^{-1}(\exp(\Delta A) - I)\Delta B. \tag{8}$$

In this context, $\Delta$ represents the step size. Using $h$ and $x$ to denote the state vector and the input vector, respectively, we can derive the following expression similar to that of recurrent neural networks (RNNs):

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t \tag{9}$$
$$y_t = Ch_t. \tag{10}$$

The relationship between $h_t$ and $y_t$ can be discovered through multistep iterations. The expressions for the first three terms are listed below

$$h_0 = \overline{B}x_0 \tag{11}$$
$$y_0 = C\overline{B}x_0 \tag{12}$$
$$h_1 = \overline{AB}x_0 + \overline{B}x_1 \tag{13}$$
$$y_1 = C\overline{AB}x_0 + C\overline{B}x_1 \tag{14}$$
$$h_2 = \overline{A}^2\overline{B}x_0 + \overline{AB}x_1 + \overline{B}x_2 \tag{15}$$
$$y_2 = C\overline{A}^2\overline{B}x_0 + C\overline{AB}x_1 + C\overline{B}x_2. \tag{16}$$

The transformation relationship can be generalized to any point in time as follows:

$$y_m = (x_0, x_1, x_2, \ldots, x_{m-1}) \otimes \begin{pmatrix} C\overline{B} \\ C\overline{AB} \\ C\overline{A}^2\overline{B} \\ \vdots \\ C\overline{A}^{m-1}\overline{B} \end{pmatrix} \tag{17}$$

where $m$ is the length of the input sequence, and $\otimes$ denotes the convolution operation. The state-of-the-art SSM model, Mamba, further refines $B$, $C$, and $A$, making it dependent on the input, thus implementing dynamic feature representation. Mamba is developed on the basis of the S4 model, with a recursive form similar to (9), and can sense long sequences and activate more neurons to assist in fusion. In addition, Mamba adopts a scanning method with parallel processing advantages similar to (17), thereby improving efficiency.

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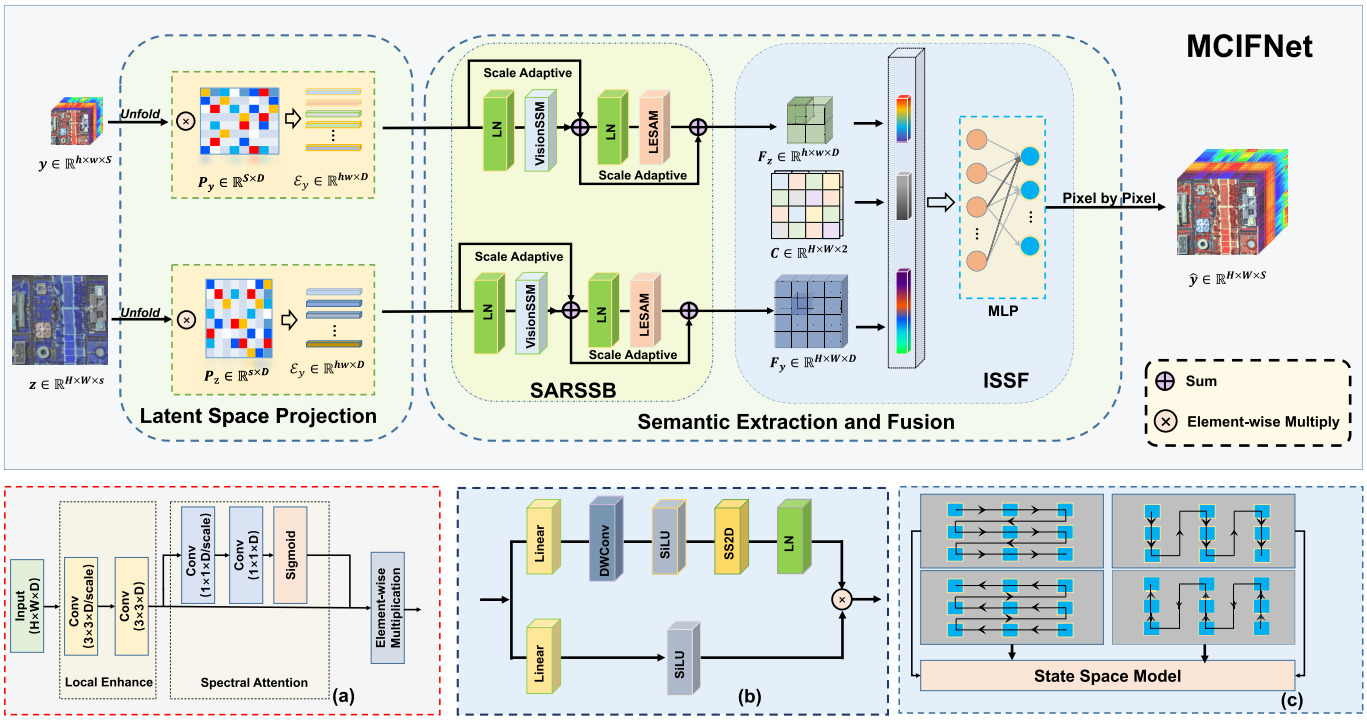



Fig. 2. Network of MCIFNet. (a) LESAM. (b) VisionSSM. (c) SS2D.

### B. Network Design

The proposed MCIFNet (Fig. 2) comprises two core units: Latent space projection and semantic extraction and fusion. The workflow of MCIFNet is as follows: First, the HSI and MSI are separately fed into the LSPU for shallow encoding and projection into the latent space, providing encoded shallow features for subsequent semantic extraction and fusion. Subsequently, the projected HSI and MSI are input into the SEFU for semantic feature extraction and point-to-point feature fusion in the continuous domain, ultimately generating the fused image.

*1) Local Enhancement and Spectral Attention Module:* The LESAM was constructed inspired by SENet [52] to compensate for the lack of local information in the Mamba output features and to enhance the expressiveness of the spectral dimension. LESAM [Fig. 2(a)] adopts the two-stage strategy of local enhancement and spectral attention. Local enhancement achieves feature channel compression and reactivation through a pair of $3 \times 3$ convolutional layers, where the primary convolution effectively reduces channel redundancy, and the secondary convolution recovers and enhances local feature nuances through an excitation mechanism, which improves the local perceptual ability of the network. Subsequently, LESAM is incorporated into a set of $1 \times 1$ convolutional layers to perform two tasks: first, channel dimension squeezing, and then excitation via a second $1 \times 1$ convolutional layer. This process takes advantage of the intrinsic correlation between spectral features to optimize and refine the spectral feature representations, which not only improves the validity of the features but also enhances the model's understanding of the complex interactions of the spectral features. Ultimately, the refined features are fed into a Sigmoid activation layer that generates attention weights specific to each channel. These weights are elementwise multiplication with the previously enhanced local features to modulate the contribution of each channel and achieve synergistic enhancement of local details and spectral feature expression. The LESAM is represented as

$$\mathcal{B} = \mathrm{Conv}_{3D}(\mathrm{Conv}_{3\mathrm{Ds}}(\mathrm{Inp})) \tag{18}$$

$$\mathcal{A} = \mathrm{Sigmoid}(\mathrm{Conv}_{1D}(\mathrm{Conv}_{1\mathrm{Ds}}(\mathcal{B}))) \times \mathcal{B} \tag{19}$$

where Inp is the input feature, $\mathrm{Conv}_{3D}$ is a convolution kernel of size $3 \times 3$ with kernel dimension $D$, $s$ is the channel compression ratio, $\mathcal{B}$ is the local enhanced features, and $\mathcal{A}$ is the enhanced spectral features.

*2) Vision SSM:* The VisionSSM model [Fig. 2(b)] uses a dual-path architecture to enhance feature representation. Path 1 is designed to extract both local and global interaction information. It begins by extending the input in the channel dimension through a linear layer, followed by depthwise convolution and SiLU activation for enhanced local feature representation. The SS2D mechanism is then applied to capture long-range dependencies, enriching the global context. A normalization layer is included to stabilize training and accelerate convergence. Path 2 focuses on preserving the original feature information. It maps the input to the same dimension as Path 1 using a linear layer and SiLU activation, preserving the original feature information and reducing information loss. Finally, the outputs from both the paths are combined using the Hadamard product to generate a more robust and comprehensive feature representation. The formula is expressed as

$$f = \mathrm{Linear}(\mathrm{Inp}) \tag{20}$$

$$F = \mathrm{LN}(\mathrm{SS2D}(\mathrm{SiLU}(\mathrm{DWConv}(f)))) \times \mathrm{SiLU}(f) \tag{21}$$

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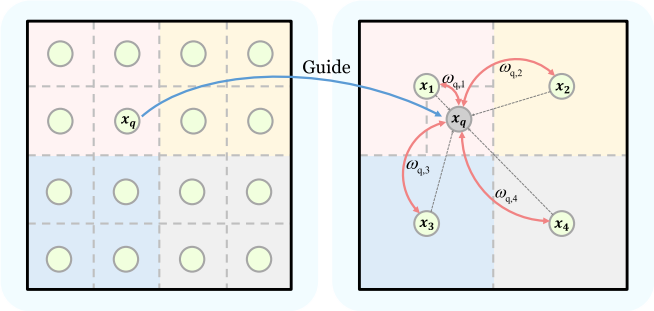



Fig. 3. Diagram of ISSF. The spectral information of the target query point $x_q$ is predicted by a neural network using the high-resolution spectral vectors of the query point $x_q$ and combining the four adjacent spectra in the low-resolution image. The fusion features are generated by traversing all the query points.

where the process of SS2D [Fig. 2(c)] is: first, the input features are expanded into sequences seq $=$ $\{\text{seq}_1, \text{seq}_2, \text{seq}_3, \text{seq}_4\}$. For each sequence $\text{seq}_i$, a specific SSM is used for feature extraction, and finally, global feature sensing is achieved by additive fusion, i.e.,

$$\mathcal{S}\text{out} = \sum_{i=1}^{4} \mathbf{SSM}_i(\mathbf{seq}_i) \tag{22}$$

where $\mathbf{SSM}_i$ is the SSM used for the $i$th sequence.

*3) ISSF Module:* The ISSF module performs upsampling of low spatial resolution features ($lr \in \mathbb{R}^{h\times w\times C}$) guided by high spatial resolution features ($hr \in \mathbb{R}^{H\times W\times c}$) in the continuous domain to achieve efficient spatial–spectral fusion, as illustrated in Fig. 3. The specific process is as follows: first, geometric alignment between the $hr$ and $lr$ features is achieved using a grid, followed by spectral interpolation at any queried pixel location. The interpolation process can be described as: for each query point, we first obtain the set of four neighboring spectral vectors from the $lr$ feature map, $N_q = \{x_i | i = 1, 2, 3, 4\}$, as well as the corresponding spectral vector at the same position in the $hr$ feature map. Next, a multilayer perceptron (MLP) is used to fit the spectral vector of the query point. Specifically, for each neighboring pixel $x_i$ in $lr$, the spectral vector value and interpolation weight for the query point are computed as follows:

$$v_{(q,i)}, w_{(q,i)} = f_\theta(lr(x_i), hr(x_q), x_q - x_i) \tag{23}$$

where $v_{(q,i)} \in \mathbb{R}^{1\times 1\times C}$ represents the spectral vector of the query point, and $w_{(q,i)}$ is the interpolation weight at point $x_i$ in the $lr$ feature map. $f_\theta$ represents the MLP with learnable parameters $\theta$. $x_q - x_i$ are the relative coordinates. The final spectral vector $v$ for the query point $x_q$ is then computed using the following equation:

$$v = \sum_{i=1}^{4} v_{(q,i)} \times \text{softmax}(w_{(q,i)}). \tag{24}$$

By applying this process to all the query points, spatial and spectral fusion between the $hr$ and $lr$ features is achieved.

*4) Latent Space Projection Unit:* The LSPU aims to shallowly encode and project HSI and MSI into a latent feature space. The HSI ($y \in \mathbb{R}^{h\times w\times S}$) and MSI ($z \in \mathbb{R}^{H\times W\times s}$) are first unfolded, and then mapped to the same spectral dimensions by the projection matrices ($P_y \in \mathbb{R}^{S\times D}$) and ($P_z \in \mathbb{R}^{s\times D}$), respectively, to generate the corresponding feature representations ($\mathcal{E}_y \in \mathbb{R}^{hw\times D}$) and ($\mathcal{E}_z \in \mathbb{R}^{HW\times D}$). Specifically, to enhance the semantic representation of the latent space, we adopt the local assemble strategy, which scans the image by setting the sliding window of size $k \times k$ on the image, performs unfold operation on the scanned region, and then aggregates the information to the channel dimension by the projection matrix, which not only preserves the detailed features of the local spatial structure but also facilitates the interactions between the feature channels. The process can be expressed as

$$\mathcal{E}_y = \text{unfold}(y) \times P_y \tag{25}$$

$$\mathcal{E}_z = \text{unfold}(z) \times P_z \tag{26}$$

where $k$ and $D$ were set to 3 and 64, respectively.

*5) Semantic Extraction and Fusion Unit:* The SEFU is designed with two main components: SARSSB and ISSF. SEFU uses SARSSB to extract information from the projected HSI and MSI using a scanning method, and then ISSF is designed to achieve the fusion of dual-mode information, finally obtaining the fused image. Specifically, in SARSSB, the projection of HSI and MSI is first unified, and the feature maps of the same track are balanced. Next, through index operation, the features are rearranged to obtain the feature maps of the same track. Then, the rearranged features are sent to the vision SSM (VSSM) for processing. The expression of SARSSB is as follows:

$$M_z = \alpha * \varepsilon_z + \text{VSSM}(\text{LN}(\mathcal{E}_z)) \tag{27}$$

$$F_z = \beta * M_z + \text{LESAM}(\text{LN}(M_z)) \tag{28}$$

$$M_y = \gamma * \varepsilon_y + \text{VSSM}\big(\text{LN}(\mathcal{E}_y)\big) \tag{29}$$

$$F_y = \delta * M_y + \text{LESAM}\big(\text{LN}(M_y)\big) \tag{30}$$

where LN is the layer normalization, VSSM is the vision SSM, and $M_z \in \mathbb{R}^{H\times W\times D}$ and $M_y \in \mathbb{R}^{h\times w\times D}$ are the intermediate states of the SEFU output for MSI and HSI, respectively. LESAM is used to compensate for local features and promote spectral representation capabilities. $\alpha$, $\beta$, $\gamma$, and $\delta$ are learnable adaptive parameters used to control the connection information. According to the global average pooling of SARSSB, the deep features $F_z \in \mathbb{R}^{H\times W\times D}$ and $F_y \in \mathbb{R}^{h\times w\times D}$ of MSI and HSI are generated. Next, we use the ISSF to make point-by-point predictions generate the fused image

$$\hat{y} = \text{ISSF}(F_z, F_y) \tag{31}$$

where $\hat{y} \in \mathbb{R}^{H\times W\times C}$ is the final fused image.

## C. Loss Function

In this study, L1 loss is selected to measure the gap between the fused and reference images, which is due to the fact that

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TABLE I
SPECIFICS OF FOUR DATASETS

| Datasets | Chikusei | DFC2018 | PaviaU | PaviaC |
|---|---|---|---|---|
| Number of bands | 128 | 48 | 103 | 102 |
| Pixel resolution (m) | 2.5 | 1 | 1.3 | 1.3 |
| Spectral range (nm) | 343-1018 | 380-1050 | 430-860 | 430-860 |
| Size | $2354 \times 2185$ | $1202 \times 4172$ | $610 \times 340$ | $1096 \times 1096$ |

the traditional L2 loss penalizes large and small errors differently [53], resulting in insufficient fidelity of high-frequency information. The mathematical expression is as follows:

$$\mathcal{L}\text{oss} = \frac{\sum_{i=1}^{n} |\text{Out}_i - \text{Ref}_i|}{n} \tag{32}$$

where Out is the output of the current network, Ref is the reference image, and $n$ is the pixel index.

## IV. EXPERIMENTS

### A. Datasets

This study uses datasets from four different sources, Chikusei, DFC2018, PaviaC, and PaviaU, to comprehensively evaluate the performance of MCIFNet in diverse environments. These datasets have different spatial and spectral resolutions, which enable the comprehensive validation of the fusion of the algorithms under different data distributions. The Chikusei dataset was acquired on 29 July 2014 with the Headwall Hyperspec-VNIR-C sensor and contains 19 feature classes. The DFC2018 dataset is derived from the ITRES CASI 1500 sensor, which is open source for the 2018 IEEE GRSS Data Fusion Competition, provided by Dr Saurabh Prasad and contains 20 identified feature classes. The PaviaC and PaviaU datasets were captured in Northern Italy using the ROSIS sensor, carefully processed to discard certain discarded bands and contain nine feature classes each. By conducting experimental analyses on these datasets, the applicability of MCIFNet in various environmental conditions can be effectively evaluated. The specific dataset parameters are listed in Table I.

### B. Implementation Details

The experimental setup for this study relies on the Python 3.10 environment and PyTorch 1.8, with hardware support from the NVIDIA 4090 GPU and 128G of memory. For data simulation, we follow Wald's protocol [54], using the original HSI as the reference image. The HSI and MSI samples required for training and testing are generated by spatially and spectrally downsampling the original HSI. Spectral downsampling is simulated using the spectral response function of the Gaofen-2 satellite. For spatial downsampling, we use Gaussian filtering and bilinear interpolation to simulate realistic image degradation, ensuring the credibility of the simulation results. For each dataset, we crop a $192 \times 192$ region as the test set, while the remaining part is used as the training set. It is important to note that there is no overlap between the test set and the training set, which helps more accurately evaluate the generalization capability of the model. Furthermore, to ensure fairness and comparability in training, we use the same hyperparameters and data across all deep learning algorithms. Specifically, for the training set, we set the input sample size to $128 \times 128$ and the total number of epochs to 10 000, and each epoch contains one training sample. The learning rate is set to 0.0001, and the optimizer is Adam.

### C. Evaluation Metrics

In this study, we selected root mean square error (RMSE), peak signal-to-noise ratio (PSNR), Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS), and spectral angle mapper (SAM) as evaluation metrics to comprehensively verify the fusion quality of various algorithms. The combined evaluation of these metrics helps thoroughly analyze the advantages and disadvantages of different fusion methods in terms of image quality and further validate the performance of MCIFNet in HMIF.

RMSE measures the degree of closeness between a fused image and a reference image by calculating the average difference between corresponding pixel values. A smaller RMSE value indicates that the fused image is closer to the reference image, resulting in better fusion quality. The expression is as follows:

$$\text{RMSE}(\mathcal{X}, \mathcal{Y}) = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (\mathcal{X}_i - \mathcal{Y}_i)^2} \tag{33}$$

where $\mathcal{X}$ and $\mathcal{Y}$ are the fused and reference images, respectively. $N$ is the pixel index.

PSNR measures the reconstruction quality of the fused image by assessing the ratio between signal strength and noise. The higher the value of PSNR, the better the reconstruction quality. The mathematical representation is as follows:

$$\text{PSNR}(\mathcal{X}, \mathcal{Y}) = \frac{1}{B} \sum_{i=1}^{B} 10 \cdot \log_{10} \left( \frac{\max(\mathcal{Y}_i)^2}{\text{mse}(\mathcal{X}_i, \mathcal{Y}_i)} \right) \tag{34}$$

where $B$ is the number of bands, and mse is the mean square error.

ERGAS comprehensively considers mse, RMSE, and brightness information to evaluate the spatial and spectral balance of fused images. A smaller ERGAS value indicates higher quality of the fused image. The formula is as follows:

$$\text{ERGAS}(\mathcal{X}, \mathcal{Y}) = \frac{100}{r} \sqrt{\frac{1}{B} \sum_{i=1}^{B} \left( \frac{\text{RMSE}(\mathcal{X}_i, \mathcal{Y}_i)}{\overline{\mathcal{Y}_i}} \right)^2}. \tag{35}$$

SAM calculates the spectral difference between the reference image and the fused image by measuring the angle between their vectors in the spectral space. A lower SAM

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TABLE II
EXPERIMENTAL RESULTS ON CHIKUSEI

| Ratio | Metrics | GSA | CNMF | HySure | LEMamba | DCT | MoGDCN | TSFN | 3DT-Net | MCIFNet |
|---|---|---|---|---|---|---|---|---|---|---|
| ×4 | RMSE | **0.5213** | 0.5691 | 0.5338 | 0.6545 | 0.7347 | 0.6578 | **0.4685** | 0.5877 | **0.4660** |
| | PSNR | **44.8028** | 44.0415 | 44.5975 | 42.8266 | 41.8226 | 42.7831 | **45.7305** | 43.7617 | **45.7766** |
| | ERGAS | **1.8500** | 2.0615 | 2.0789 | 2.3247 | 2.1720 | 2.4320 | **1.8899** | 2.3701 | **1.9913** |
| | SAM | **0.8974** | 0.9134 | 0.9161 | 1.1903 | 1.0163 | 1.0737 | **0.8133** | 1.0238 | **0.8718** |
| ×8 | RMSE | 0.8374 | 0.9859 | 0.9962 | 0.6882 | 0.9045 | 0.5927 | **0.5108** | **0.5684** | **0.5077** |
| | PSNR | 40.6865 | 39.2682 | 39.1776 | 42.3909 | 40.0163 | 43.6887 | **44.9794** | **44.0510** | **45.0326** |
| | ERGAS | 2.4025 | 2.3773 | 2.6155 | 2.4118 | 2.5491 | 2.3349 | **2.0047** | **2.2130** | **1.9935** |
| | SAM | 1.3101 | 1.3665 | 1.5117 | 1.2423 | 1.1438 | 1.0885 | **0.9336** | **1.0245** | **0.9421** |
| ×16 | RMSE | 2.6932 | 3.3952 | 3.7335 | 0.7163 | 0.9755 | 0.6298 | **0.5892** | **0.6010** | **0.5403** |
| | PSNR | 30.5392 | 28.5274 | 27.7023 | 42.0424 | 39.3601 | 43.1610 | **43.7388** | **43.5678** | **44.4926** |
| | ERGAS | 6.1269 | 9.079 | 7.9907 | 2.3545 | 2.7263 | **2.3612** | **2.1313** | 2.3983 | **2.0408** |
| | SAM | 3.7424 | 3.2738 | 5.5799 | 1.2792 | 1.3210 | 1.1521 | **1.0296** | **1.0887** | **0.9968** |

value indicates higher spectral similarity. The expression for SAM is as follows:

$$\mathrm{SAM}(\mathcal{X},\mathcal{Y}) = \frac{1}{HW}\sum_{i,j}\arccos\left(\frac{\mathcal{Y}_{ij}^{T}\cdot\mathcal{X}_{ij}}{\|\mathcal{Y}_{ij}\|_2\|\mathcal{X}_{ij}\|_2}\right) \tag{36}$$

where $H$ and $W$ are the width and height of the image, respectively.

### D. Experimental Results and Analysis

To validate the superiority of MCIFNet, in this study, pansharpening algorithm GSA [25], decomposition CNMF [55], and HySure [56]-based deep learning algorithms LEMamba [57], DCT [1], MoGDCN [46], TSFN [42], and 3DT-Net [47] were selected as the comparative methods for a series of fusion experiments. Specifically, the best, second-best and third-best results in the experiments are highlighted in red, green and blue fonts, respectively.

*1) Results on Chikusei:* In this section, we evaluate the performance of MCIFNet on the Chikusei dataset. Table II lists the performance metrics of various algorithms on this dataset, showing that MCIFNet demonstrates the best overall performance. However, in the 4× fusion, the traditional algorithm GSA performs best in terms of ERGAS, which may be due to the lower requirement for nonlinear fitting capabilities at this scale factor, allowing the traditional algorithm to effectively adapt. Nonetheless, analyses of RMSE, PSNR, and SAM indicate that MCIFNet remains a competitive choice overall. It is noteworthy that the deep learning algorithm DCT failed to converge effectively in the experiment, exhibiting local optima, possibly because the DCT model could not adapt to the specific distribution of this dataset. In the 8× scale factor fusion, TSFN achieves the best SAM, while MCIFNet performs best in the other metrics. This indicates that in this case, MCIFNet slightly lags behind TSFN in spectral fidelity, but still shows overall superiority. For the 16× fusion, where the fusion process involves a higher degree of nonlinearity, all the deep learning algorithms outperform traditional physical algorithms, mainly due to the strong nonlinear fitting capabilities of deep learning. Since the performance of the algorithms cannot be effectively distinguished through RGB display of the fused images, we further analyze by plotting MRAE and SAM residual heatmaps for the 8× fusion and the difference map of the 29th band of the fusion results (Fig. 4). The results show that MCIFNet has the smallest residuals and the fusion results are closest to the reference image, further proving the superiority of MCIFNet over other algorithms.

*2) Results on DFC2018:* Table III presents the quantitative evaluation results of various algorithms on the DFC2018 dataset, demonstrating that MCIFNet achieves significant advantages in fusion at 4×, 8×, and 16× scale factors. In addition, the heatmap and residual map in Fig. 5 further illustrate that MCIFNet exhibits smaller residuals compared with other algorithms, with fusion results that are closer to the reference image. For the 4× scale factor fusion, GSA shows competitive performance, indicating that traditional physical model-driven algorithms remain competitive for low-scale factor fusion. However, in the 8× scale factor fusion, MoGDCN and TSFN achieve suboptimal performance in two metrics each, differing from their rankings on the Chikusei dataset. This suggests that the performance of deep learning algorithms is influenced not only by the network architecture but also by the data distribution. For the 16× scale factor fusion, 3DT-Net achieves the second-best performance, which indicates that 3DT-Net is a notably competitive algorithm in this distribution for high-scale factor fusion, apart from MCIFNet. The ability of MCIFNet to efficiently extract deep features under this data distribution leads to more realistic fusion results, further validating its robustness across different scale factors. In summary, the results of MCIFNet on the DFC2018 dataset reaffirm its outstanding performance in the HMIF field and its significant advantages over traditional algorithms and other deep learning models.

*3) Results on PaviaC:* The PaviaC dataset has higher spectral resolution compared with the DFC2018 dataset but relatively lower spatial resolution, indicating the fusion process has a higher demand for nonlinear fitting. Table IV provides the quantitative evaluation of MCIFNet and comparison algorithms on the PaviaC fusion experiments, showing that MCIFNet achieves the best performance across all the metrics. This suggests that Mamba-based feature extraction and INR-based cross-modal fusion remain effective fusion strategies. In the 4× scale factor fusion, the traditional algorithm

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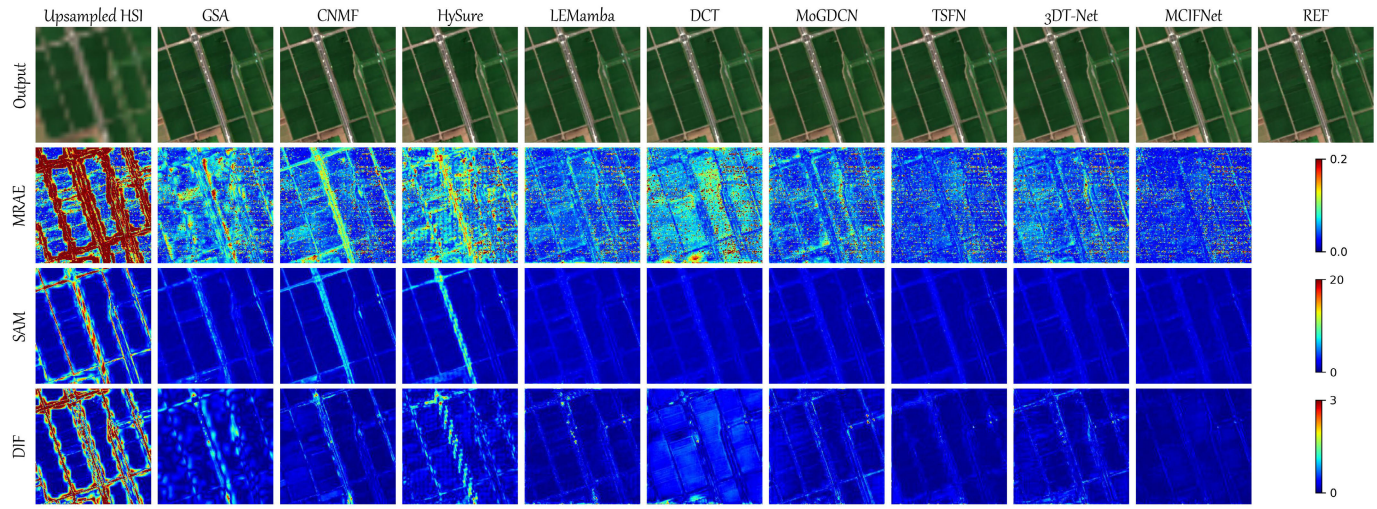



Fig. 4. Experimental results for the Chikusei dataset (the R-G-B channels correspond to the 61-41-21 bands, and the DIF indicates the difference in the 29th band).

TABLE III
EXPERIMENTAL RESULTS ON DFC2018

| Ratio | Metrics | GSA | CNMF | HySure | LEMamba | DCT | MoGDCN | TSFN | 3DT-Net | MCIFNet |
|---|---|---|---|---|---|---|---|---|---|---|
| ×4 | RMSE | **0.2697** | 0.3923 | 0.3509 | 0.4647 | 0.6291 | 0.4819 | **0.3360** | 0.3452 | **0.2683** |
| | PSNR | **46.6640** | 43.4096 | 44.3785 | 41.9381 | 39.3082 | 41.6224 | **44.7543** | 44.5201 | **46.7101** |
| | ERGAS | **0.6276** | 0.9577 | 0.9385 | 1.1760 | 1.4345 | 1.0686 | **0.7686** | 0.8080 | **0.6794** |
| | SAM | **1.2275** | 1.4295 | 1.4063 | 1.9227 | 1.7740 | 1.9012 | 1.4045 | **1.3900** | **1.2294** |
| ×8 | RMSE | 0.6468 | 0.8463 | 0.9451 | 0.4833 | 0.8515 | **0.3484** | **0.3507** | 0.3555 | **0.3096** |
| | PSNR | 39.0673 | 36.7320 | 35.7730 | 41.5986 | 36.6785 | **44.4406** | **44.3831** | 44.2654 | **45.4661** |
| | ERGAS | 1.3631 | 1.8750 | 2.1158 | 1.1887 | 1.9829 | **0.8290** | **0.8036** | 0.8461 | **0.7700** |
| | SAM | 2.5129 | 2.8498 | 3.2806 | 2.0161 | 1.8071 | **1.5020** | **1.4257** | 1.5313 | **1.3827** |
| ×16 | RMSE | 2.3920 | 3.4168 | 4.4758 | 0.4896 | 0.6794 | **0.3800** | 0.3923 | **0.3649** | **0.3244** |
| | PSNR | 27.7071 | 24.6099 | 22.2647 | 41.4849 | 38.6402 | **43.6862** | 43.4094 | **44.0395** | **45.0607** |
| | ERGAS | 5.1531 | 7.0132 | 9.3742 | 1.2023 | 1.5911 | 0.9264 | **0.9044** | **0.8842** | **0.8000** |
| | SAM | 9.6318 | 9.1078 | 18.5527 | 2.0628 | 2.1869 | 1.6649 | **1.6465** | **1.5508** | **1.4391** |

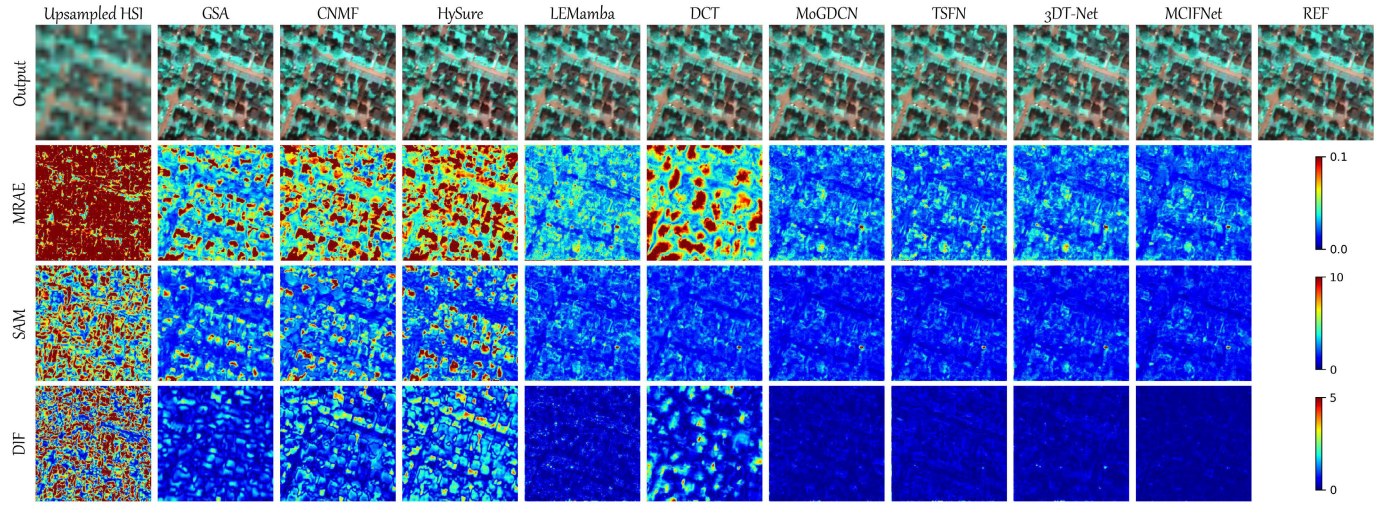



Fig. 5. Experimental results for the DFC2018 dataset (the R-G-B channels correspond to the 41-31-21 bands, and the DIF indicates the difference in the 11th band).

GSA demonstrates its strengths in ERGAS. However, the generalization ability of GSA significantly lags behind when faced with 8× and 16× scale factors. In contrast, DCT ranks lowest among all the deep learning algorithms, indicating its relatively weak ability to adapt to this remote sensing dataset. The MRAE and SAM heatmaps and difference maps in Fig. 6

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TABLE IV
EXPERIMENTAL RESULTS ON PAVIAC

| Ratio | Metrics | GSA | CNMF | HySure | LEMamba | DCT | MoGDCN | TSFN | 3DT-Net | MCIFNet |
|---|---|---|---|---|---|---|---|---|---|---|
| ×4 | RMSE | 2.1346 | 2.6844 | 2.4564 | 2.2043 | 2.9616 | **2.0575** | **2.0633** | 2.0956 | **1.8854** |
| | PSNR | 41.5443 | 39.5539 | 40.3247 | 41.2652 | 38.7002 | **41.8641** | **41.8396** | 41.7048 | **42.6227** |
| | ERGAS | **1.7756** | 2.2747 | 2.0394 | 1.8982 | 2.3605 | 1.8024 | **1.7819** | 1.8130 | **1.6684** |
| | SAM | 3.4226 | 3.2693 | 3.4089 | 3.3781 | 3.3969 | 3.2585 | **3.1132** | **3.2519** | **3.0767** |
| ×8 | RMSE | 3.1770 | 4.0152 | 4.2827 | 2.2905 | 3.8059 | 2.1528 | **2.1323** | **2.0992** | **1.8982** |
| | PSNR | 38.0903 | 36.0567 | 35.4964 | 40.9320 | 36.5217 | 41.4706 | **41.5537** | **41.6897** | **42.5638** |
| | ERGAS | 2.4695 | 3.1181 | 3.2933 | 1.9168 | 2.9364 | 1.8786 | **1.8509** | **1.8263** | **1.6854** |
| | SAM | 4.8678 | 4.2914 | 5.5204 | 3.4364 | 3.8152 | 3.4145 | **3.2271** | **3.3018** | **3.0958** |
| ×16 | RMSE | 7.3343 | 10.1476 | 10.0001 | 2.2863 | 4.8329 | **2.1769** | 2.3545 | **2.2285** | **1.9227** |
| | PSNR | 30.8236 | 28.0035 | 28.1308 | 40.9482 | 34.4466 | **41.3742** | 40.6929 | **41.1706** | **42.4524** |
| | ERGAS | 5.0613 | 7.0045 | 6.6657 | 1.9119 | 3.6603 | **1.8754** | 1.9634 | **1.8982** | **1.6825** |
| | SAM | 12.8790 | 9.0045 | 14.4382 | 3.4526 | 3.6635 | **3.4384** | 3.4836 | **3.4450** | **3.1017** |

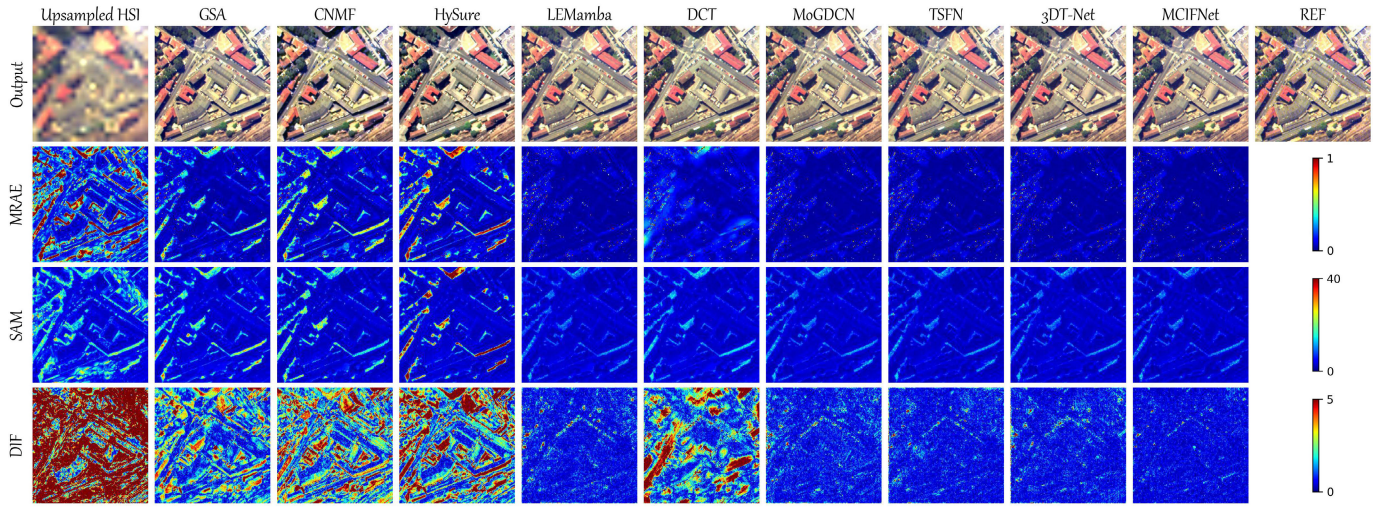



Fig. 6. Experimental results for the PaviaC dataset (the R-G-B channels correspond to the 61-41-21 bands, and the DIF indicates the difference in the 31st band).

also show that MCIFNet has the lightest color, indicating that its fusion results are closer to the reference image. This further confirms that deep learning methods significantly outperform physics-based models in handling large-scale factor fusions.

*4) Results on PaviaU:* This section discusses the comparative analysis of MCIFNet and other HMIF algorithms on the PaviaU dataset. Compared with the PaviaC dataset, the PaviaU dataset has more bands, making data fusion more challenging. Table V clearly shows that MCIFNet outperforms all other methods across all the evaluation metrics, indicating that the Mamba combined with the INR approach effectively extracts deep features, resulting in high-quality data fusion. Observing the MRAE heatmap and the residual map of the 41st band, the advantages of MCIFNet are evident. Although the differences in the SAM heatmap are less visually apparent, careful inspection reveals that MCIFNet has the smallest residuals. In this data distribution, all the deep learning methods perform well. Both the quantitative evaluation and the heatmaps and residual maps (Fig. 7) demonstrate that deep learning methods can more deeply perceive effective mappings between data. These results further confirm the advantages of deep learning in HMIF and again reiterate that while traditional algorithms perform relatively well for small scale factor HMIF, they may not meet practical application requirements for larger scale factors such as 8× or 16×.

### E. Comparison of Spectral Curves

This section presents the spectral curves of the 8× scale factor fusion results for the DFC2018 dataset, aiming to evaluate the spectral fidelity performance of different algorithms by comparing the fusion results to the ground truth. Fig. 8 clearly shows that the spectral curve of MCIFNet is closer to the real curves compared with other algorithms, indicating superior spectral fidelity. This suggests that MCIFNet can provide more reliable data support for subsequent practical applications that rely on accurate spectral information, such as land cover classification and environmental monitoring.

### F. Band-by-Band Comparison of PSNR

While previous experiments evaluate the overall fusion results, this section focuses on the detailed PSNR statistics of each band for the 8× scale factor fusion results on the DFC2018 dataset to validate the superiority of MCIFNet. Fig. 9 shows that although MCIFNet does not achieve the best performance for every band, it still demonstrates exceptional

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TABLE V
EXPERIMENTAL RESULTS ON PAVIAU

| Ratio | Metrics | GSA | CNMF | HySure | LEMamba | DCT | MoGDCN | TSFN | 3DT-Net | MCIFNet |
|---|---|---|---|---|---|---|---|---|---|---|
| ×4 | RMSE | 3.1551 | 3.3015 | 3.3610 | 2.7690 | 3.3989 | 2.7693 | **2.5856** | **2.6153** | **2.2627** |
| | PSNR | 38.1505 | 37.7567 | 37.6013 | 39.2843 | 37.5041 | 39.2833 | **39.8796** | **39.7803** | **41.0383** |
| | ERGAS | 2.0478 | 2.3140 | 2.2900 | 1.9930 | 2.3185 | 2.0454 | **1.9328** | **1.9633** | **1.7196** |
| | SAM | 3.1960 | 3.0233 | 3.1485 | 2.2703 | 2.6767 | 2.8935 | **2.4508** | **2.5797** | **2.2849** |
| ×8 | RMSE | 4.6438 | 3.2554 | 5.9694 | 2.8785 | 4.4947 | **2.6381** | 2.7623 | **2.6852** | **2.2999** |
| | PSNR | 34.7934 | 37.8788 | 32.6122 | 38.9474 | 35.0768 | **39.7051** | 39.3052 | **39.5511** | **40.8966** |
| | ERGAS | 2.7986 | 2.2497 | 3.6370 | 2.0740 | 3.0057 | **1.9859** | 2.0247 | **2.0060** | **1.7373** |
| | SAM | 5.3672 | 3.0220 | 5.9395 | 2.8558 | 2.8949 | 2.6406 | **2.6047** | **2.6028** | **2.3237** |
| ×16 | RMSE | 9.6969 | 15.6085 | 13.6783 | 2.8963 | 5.6922 | **2.7281** | 3.2029 | **2.8055** | **2.3052** |
| | PSNR | 28.3981 | 24.2636 | 25.4101 | 38.8940 | 33.0251 | **39.4135** | 38.0200 | **39.1706** | **40.8767** |
| | ERGAS | 6.2702 | 9.8748 | 7.9000 | 2.0488 | 3.6343 | **2.0568** | 2.3285 | **2.0784** | **1.7410** |
| | SAM | 10.5225 | 8.2189 | 13.7358 | 2.8685 | 3.0459 | **2.7407** | 2.8918 | **2.7896** | **2.3359** |

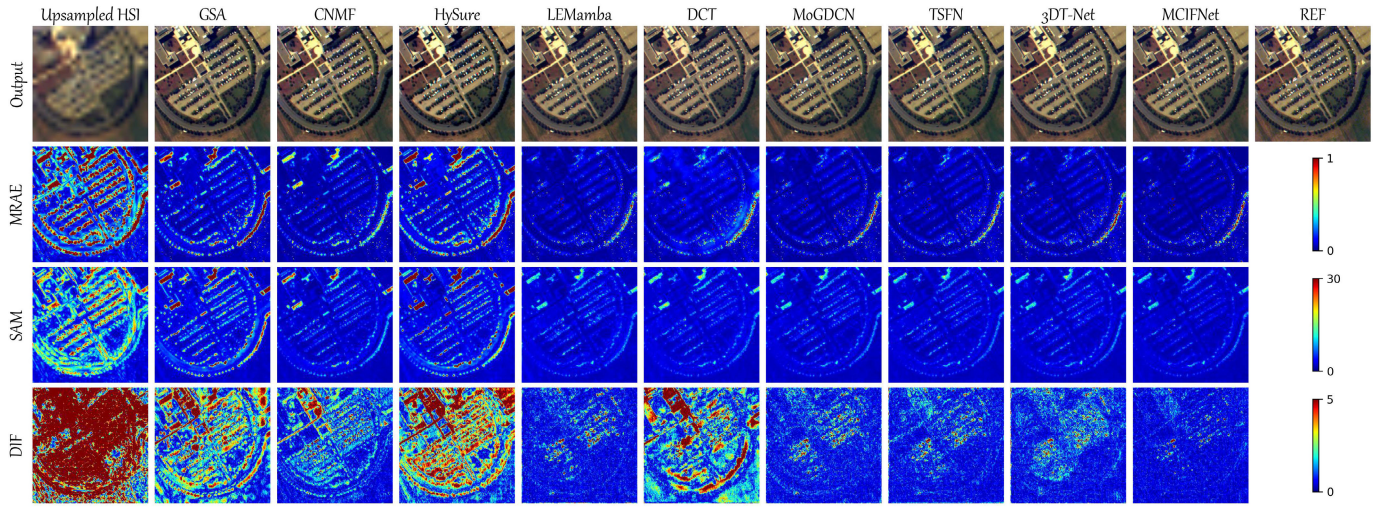



Fig. 7. Experimental results for the PaviaU dataset (the R-G-B channels correspond to the 61-51-21 bands, and the DIF indicates the difference in the 41st band).

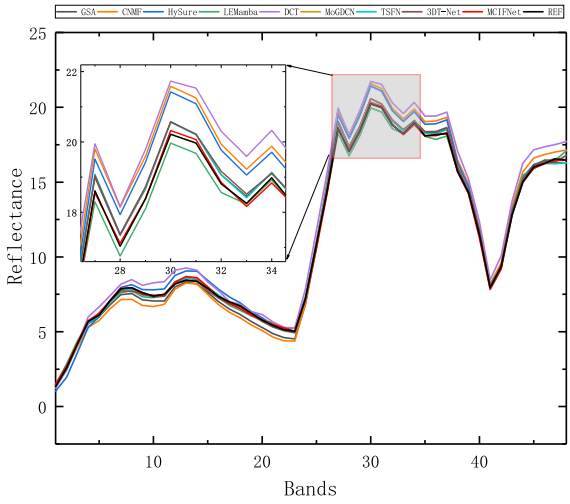



Fig. 8. Spectral curve comparison.

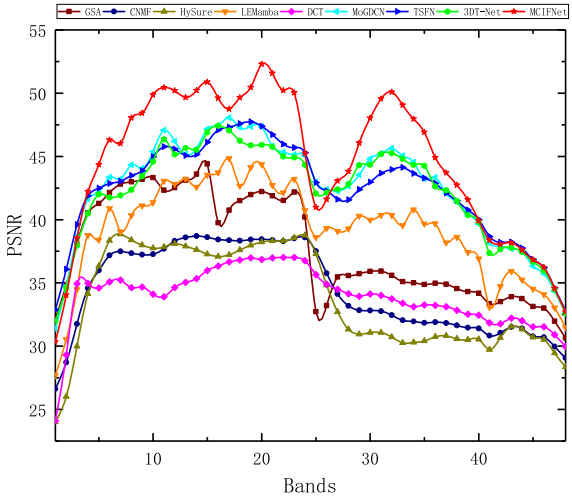



Fig. 9. PSNR curve comparison.

performance on most bands. This is because the training strategy of MCIFNet emphasizes global optimization, aiming to capture and enhance the complex interband relationships to achieve an optimal comprehensive solution. This finding also highlights the effectiveness of the Mamba combined with the INR architecture, which promotes the learning and integration of deep features from the input data. In summary, MCIFNet not only exhibits excellent reconstruction quality at

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TABLE VI
QUANTITATIVE EVALUATION OF ABLATION EXPERIMENTS

| Methods | RMSE | PSNR | ERGAS | SAM |
|---|---|---|---|---|
| w/o ISSF | 0.2810 | 46.3082 | 0.6951 | 1.2912 |
| w/o LESAM | 0.2912 | 45.9970 | 0.7314 | 1.3091 |
| w/o SARSSB | 0.3196 | 45.1888 | 0.7761 | 1.3594 |
| w/o SEFU | 0.3044 | 45.6134 | 0.7320 | 1.3428 |
| w/o VSSM | 0.2938 | 45.9206 | 0.7280 | 1.3209 |
| MCIFNet | **0.2683** | **46.7101** | **0.6794** | **1.2294** |

the overall level but also demonstrates fine control in the per-band analysis.

### G. Ablation Experiment

To deeply investigate the functional importance of each key module of MCIFNet and their contributions to network performance, this section designs a series of ablation experiments in the DFC2018 dataset with the fusion scale factor of 8, focusing on five core components of the network: ISSF, LESAM, SARSSB, SEFU, and VSSM. These experiments aim to reveal how each module independently and synergistically enhances the performance of MCIFNet on HMIF tasks. The experimental results are listed in Table VI, providing the basis for quantitative analysis.

*1) Validation of the Effectiveness of the ISSF:* In the field of high-dimensional multimodal feature fusion, traditional methods tend to use simple feature addition or concatenation strategies, often neglecting the differences in contribution between different feature dimensions. MCIFNet integrates the ISSF to achieve a more intelligent feature fusion mechanism. In the ablation experiments, we replaced the ISSF with the classical feature addition operation and adjusted the output channel number through subsequent convolution layers to match the HSI. As shown in Table VI, this substitution resulted in a significant decline in model performance. This comparison clearly indicates that the ISSF module can more finely balance and integrate multisource feature information. Through its enhanced nonlinear fusion capability, the ISSF module effectively improves the quality of feature representation.

*2) Indispensability of LESAM:* Given the unequal importance of information across spectral bands and the need to enhance local perception of the network, MCIFNet incorporates LESAM to optimize feature processing. By removing this module and comparing the experimental results, we observe a notable drop in accuracy. This indicates the crucial role of LESAM in guiding the network to focus on the most critical spectral features and enhancing local information extraction capabilities, thereby significantly improving the generalization performance of MCIFNet.

*3) Core Value of the SARSSB:* As a critical step in deep feature extraction, SARSSB leverages the Mamba architecture to generate more representative feature representations. In the ablation study, we deleted the SARSSB, relying solely on the LSPU combined with ISSF for fusion. The experimental results show a significant decline in accuracy, indicating that the Mamba-driven SARSSB is essential for constructing high-quality feature representations at the front end, thereby enhancing the ability of the network to extract effective features.

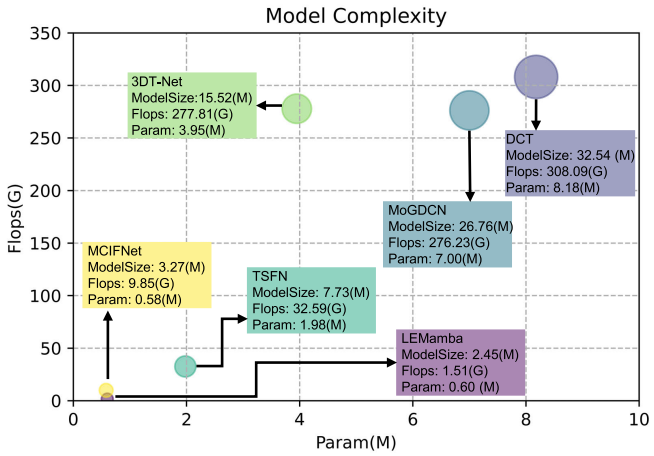



Fig. 10. Complexity statistics.

*4) Significance of VSSM:* The VSSM module plays a central role in the fusion process of MCIFNet. Its removal in ablation studies, as reflected in Table VI, resulted in a substantial drop in fusion accuracy, emphasizing the decisive role of VSSM in preserving the detail and quality of fused images. VSSM is therefore an essential component of MCIFNet, critical for achieving high-quality output.

*5) Importance of SEFU:* The SEFU is a key unit in MCIFNet, crucial for feature enhancement and fusion. To assess its importance, SEFU was removed from the model, and bilinear interpolation was used to align HSI and MSI features, followed by dimensional mapping through convolutional layers. The results in Table VI show a clear reduction in image quality, indicating that SEFU is vital for maintaining the detail and quality of the fused images.

In summary, the various components of MCIFNet work collaboratively to extract more effective deep features without causing any redundancy. This further validates the effectiveness and reliability of MCIFNet in multimodal feature fusion tasks.

### H. Complexity Comparison

In this section, we perform ModelSize, FLOPs, and Param statistics of deep learning algorithms (Fig. 10). ModelSize refers to the memory space occupied by the model, FLOPs represent the number of floating-point operations executed during the forward pass model, and Param indicates the number of learnable parameters in the model.

The results indicate that MCIFNet ranks second in terms of ModelSize and FLOPs, with a minimal gap compared with the top-ranked LEMamba. Notably, MCIFNet achieves the lowest parameter count among all the compared methods. Combined with the previous experimental results, this demonstrates that MCIFNet effectively balances fusion accuracy and lightweight design. This outcome is attributed to MCIFNet's use of the Mamba architecture and INR for a more efficient and compact network structure. These design choices effectively reduce

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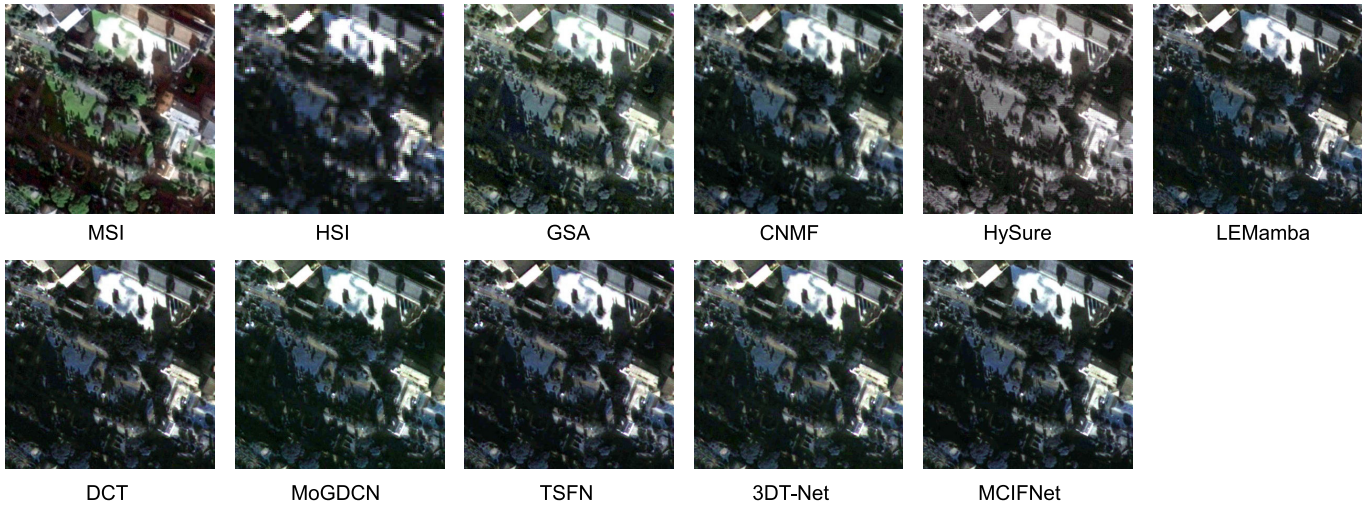



Fig. 11. Experimental results of real data.

both parameter count and computational complexity, allowing MCIFNet to maintain high performance while optimizing resource utilization. Consequently, MCIFNet offers a viable option for deployment in resource-constrained application scenarios, highlighting its advantage in computational efficiency and resource efficiency.

### I. Experiments on Real Data

In this study, we selected a three-band image with a spatial resolution of 0.5 m as MSI and an eight-band image with a resolution of 2 m as HSI. To generate the training data, we performed spatial degradation on both the HSI and MSI, using the original HSI as the reference image. Specifically, we randomly cropped 1537 sets of samples, each with a size of $128 \times 128$ pixels, for training and validating the model. The $400 \times 400$ pixel area was selected for testing. We assessed the quality of the fused images using no-reference evaluation metrics including QNR, $D_\lambda$, and $D_s$. QNR measures the combined spatial and spectral quality, $D_\lambda$ evaluates the spectral distortion of the fused image relative to the original HSI, and $D_s$ measures the spatial distortion of the fused image relative to the original MSI. It is important to note that these evaluation metrics are based on the fusion of multiple bands into a single band. Therefore, we divided the fused images into multiple groups in a many-to-one band format and used the average of these groups as the final evaluation result. This methodology ensures a comprehensive and accurate assessment of the fused image quality, maintaining high fidelity and consistency in both the spatial and spectral dimensions.

Table VII shows the quantitative evaluation of different algorithms on real data fusion, showing that MCIFNet achieves leading results on QNR, $D_\lambda$, and $D_s$, which further validates its superiority in fusion tasks. Fig. 11 shows the fusion results of different algorithms. As can be seen, the HySure and GSA algorithms show significant spectral distortions that affect the spectral consistency of the images. Although the other algorithms show no significant distortion phenomenon in visual perception, the quantitative evaluation metrics can still verify that MCIFNet has the best performance in fusion quality.

TABLE VII
QUANTITATIVE EVALUATION OF REAL DATA

| Methods | QNR | $D_\lambda$ | $D_s$ |
|---|---|---|---|
| GSA | 0.9539 | 0.0162 | 0.0304 |
| CNMF | 0.9824 | 0.0078 | 0.0099 |
| HySure | 0.8577 | 0.0694 | 0.0766 |
| LEMamba | 0.9763 | 0.0162 | 0.0076 |
| DCT | 0.9915 | 0.0043 | 0.0042 |
| MoGDCN | 0.9884 | 0.0069 | 0.0047 |
| TSFN | 0.9946 | 0.0036 | 0.0017 |
| 3DT-Net | 0.9855 | 0.0071 | 0.0074 |
| MCIFNet | **0.9977** | **0.0012** | **0.0011** |

## V. CONCLUSION

This study introduces an innovative approach to HMIF, effectively balancing global perception and computational efficiency. By leveraging the strengths of Mamba for long-range modeling and implicit neural representation for continuous domain processing, MCIFNet achieves superior performance in spatial and spectral detail reconstruction. The unique architecture of network, comprising potential space projection and SEFU s, allows for high-quality, point-to-point image fusion. The experimental results demonstrate its effectiveness across various scale factors, highlighting its potential to enhance remote sensing applications. Future work will focus on further enhancing the fusion quality while improving the efficiency of algorithm, aiming to facilitate its deployment on portable devices.

## DECLARATION OF COMPETING INTEREST

No potential conflict of interest was reported by the authors.

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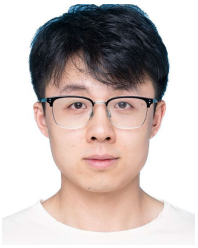

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